Practice Exercise 1

Exercise 1: Labor Productivity Growth

Scenario: A factory introduces advanced robotics in its production line, reducing human labor requirements but increasing overall output. Before automation, 100 workers produced 10,000 units per year. After automation, the factory produces 25,000 units annually with only 50 workers.

Questions:

1. Calculate labor productivity (output per worker) before and after automation.
2. What is the percentage increase in labor productivity due to automation?
3. Discuss how this improvement might affect wages and employment in the factory.

Exercise 2: The Impact of Automation on Employment

Scenario: In an industry, technological advancements reduce the labor required for a task. The labor demand function for workers is given by:

$$L = 100 - 2w$$

where L is the number of workers employed, and w is the hourly wage.

With automation, the demand function changes to:

$$L = 80 - 2w$$

Questions:

1. If the initial wage is \$20/hour, how many workers are employed before and after automation?
2. Determine the wage at which no workers are employed before and after automation.
3. Discuss the implications of this change for workers and the firm.

Exercise 3: Skill Premium and Automation

Scenario: A firm adopts a new AI technology that complements high-skilled labor but replaces low-skilled labor. Wages for high-skilled workers rise from $30/hour to $40/hour, while wages for low-skilled workers drop from $20/hour to $15/hour.

Questions:

1. Calculate the percentage change in wages for both high-skilled and low-skilled workers.
2. If the firm employs 50 high-skilled workers and 100 low-skilled workers before automation, what is the total wage bill before and after automation?
3. Analyze how the adoption of AI affects income inequality within the firm.

Exercise 4: Technological Adoption and Output Growth

Scenario: A company invests $1 million in new automation technology, which increases its annual output from 50,000 units to 80,000 units. The cost per unit decreases from $20 to $15 due to efficiency gains.

Questions:

1. What is the percentage increase in output after adopting the technology?
2. Calculate the change in total revenue if the selling price per unit remains $25.
3. Discuss how technological adoption can impact market competitiveness and prices.

Exercise 5: Job Polarization

Scenario: In an economy, job distribution shifts due to automation. The initial job shares across sectors are as follows:

- Low-skill jobs: 40%
- Middle-skill jobs: 40%
- High-skill jobs: 20%

After automation, the shares change:

- Low-skill jobs: 45%
- Middle-skill jobs: 25%
- High-skill jobs: 30%

Questions:

1. Calculate the percentage point change in job shares for each skill level.
2. If the total labor force is 10 million, how many jobs are there in each category before and after automation?
3. Discuss how automation leads to job polarization and its implications for income inequality.

Exercise 6: Automation's Effect on GDP Growth

Scenario: A country's GDP is modeled as:

$$GDP = A \times L^{0.6} \times K^{0.4}$$

where A is the technology level, L is labor input, and K is capital input.

Initially:

- $A = 1$, $L = 1,000$, $K = 500$.

After automation:

- $A = 1.2$, $L = 800$, $K = 600$.

Questions:

1. Calculate the GDP before and after automation.
2. Determine the percentage change in GDP.
3. Analyze the role of technological advancements in GDP growth and discuss how the decrease in labor input might affect the economy.